

UNDERWATER SOURCE DETECTION AND CLASSIFICATION FOR SIGNAL-BASED SURVEILLANCE: AUDIO DATASET CURATION AND CROSS-DOMAIN EVALUATION

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ABSTRACT

Machine learning for underwater acoustics is constrained by the scarcity of publicly available labeled datasets. In contrast to air-acoustic domains, where large benchmarks enable rapid model development, underwater datasets are typically small and limited in acoustic diversity, restricting robust model training and cross-domain generalization. To help address this gap, we introduce a curated underwater audio dataset derived from an open-source maritime sound archive. The dataset contains over one thousand labeled audio segments across eight biologically and mechanically relevant acoustic classes, providing an additional resource for training models in data-limited underwater environments. Additionally, we establish a lightweight Convolutional Neural Network (CNN) baseline and propose a margin-enhanced loss with feature alignment to mitigate class confusion arising from data imbalance, acoustic similarity, and cross-domain mismatch. While the baseline achieves 96.35% in-domain accuracy, evaluation on ShipsEar reveals substantial domain shift; the proposed feature alignment improves zero-shot ship detection by 42.60%, demonstrating stronger robustness under distribution mismatch. We further release a transparent curation pipeline and reproducible benchmark to support future research on imbalance mitigation, domain adaptation, and data-efficient underwater acoustic classification.

Index Terms— Audio dataset, Acoustics, Underwater Sounds, Sound Classification, Sound Source Detection

1. INTRODUCTION

Machine learning (ML) has become a key technique for underwater acoustic analysis, supporting applications such as sensor monitoring, vessel detection, maritime surveillance, and sound source classification and localization [1, 2, 3, 4, 5, 6, 7]. Despite these advances, progress remains limited by the scarcity of publicly available underwater datasets. In contrast, air-acoustic research benefits from widely used benchmarks such as ESC50 (2,000 clips) [8], GTZAN (1,000 clips) [9], UCF101 (2,840 clips) [10], and ActivityNet (1,760 clips) [11], which have enabled rapid development of deep learning models. Underwater datasets, however, remain comparatively small and fragmented [12]. Data collection in marine

environments is difficult due to harsh conditions, specialized instrumentation, and the logistical complexity of deploying and maintaining recording systems [13]. Consequently, existing datasets often contain limited acoustic diversity and uneven class distributions. Models trained on such restricted domains frequently fail to generalize across environments [14], as variations in recording location, sensor characteristics, environmental conditions, and acoustic propagation introduce substantial domain shift. Furthermore, underwater recordings commonly include overlapping biological and anthropogenic sounds, strong background noise, and class imbalance [15]. The absence of standardized preprocessing [16], segmentation, and labeling pipelines also limits reproducibility and fair benchmarking [17]. These challenges highlight the need for curated datasets and transparent data-processing pipelines that support robust evaluation and cross-domain learning in underwater acoustic machine learning.

To address these limitations, we introduce a curated underwater audio dataset (USS8) derived from an open-source maritime sound archive [18], intended for academic and non-commercial use. The recordings were manually curated to remove unrelated content and segmented into non-overlapping 1-second clips across eight acoustic categories with structured metadata. We establish a reproducible benchmark using a lightweight Tiny-CNN [19] trained on log mel-spectrogram representations. In addition, we investigate class imbalance and inter-class acoustic similarity through a margin-enhanced loss formulation that improves separation between confusing categories. To assess robustness, we further evaluate cross-domain transfer on the ShipsEar [20] dataset, examining the impact of domain shift and the effectiveness of margin-based decision-boundary regularization. The contributions of our work are summarized as follows:

- **Baseline and Imbalance Study:** A lightweight CNN benchmark with margin-based loss analysis.
- **Cross-Domain Evaluation:** Empirical assessment of transfer performance on the ShipsEar dataset.
- **Data Preparation Pipeline:** A modular audio curation and segmentation pipeline with open-source code for preparing datasets from public audio archives¹. The pipeline enables reproduction of the curated underwa-

¹<https://github.com/qtvo93/data-pipeline-avss>

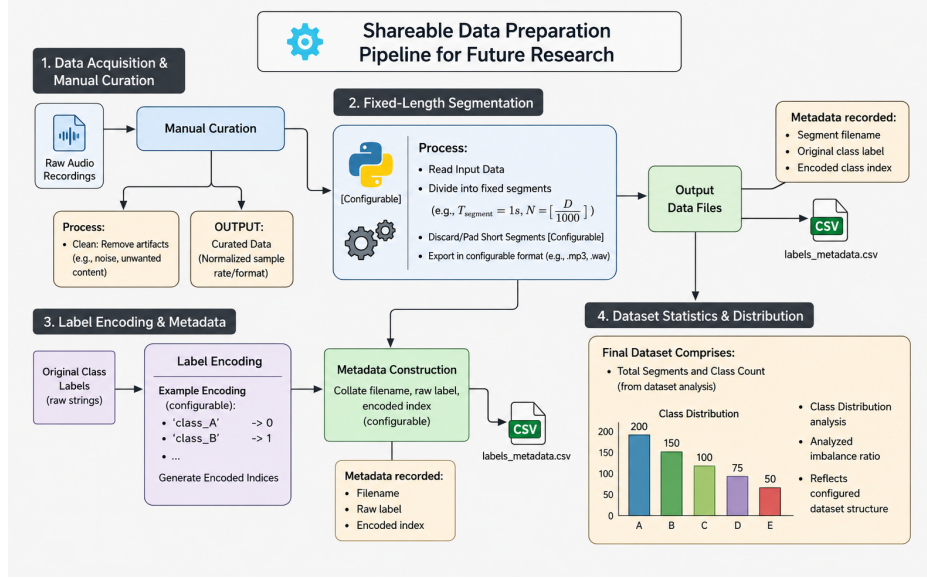


Fig. 1: Data Preparation Pipeline.

ter 8-class dataset with standardized segmentation and labeling; however, the curated dataset itself cannot be redistributed due to licensing restrictions.

2. METHODOLOGY

2.1. Audio Curation and Segmentation Procedure

2.1.1. Data Acquisition and Manual Curation

The raw audio recordings were obtained from [18]. Because the original recordings contained introductory material, silence periods, and unrelated acoustic content, each file was manually inspected prior to segmentation. Audio portions not corresponding to the labeled target events were removed to ensure strict alignment between waveform content and ground-truth annotations. This manual curation step reduces label noise and prevents contamination of the training data with irrelevant signals. Additionally, all audio files were inspected to ensure an original uniform format. This guarantees uniform spectral resolution during subsequent feature extraction and model training.

2.1.2. Fixed-Length Segmentation

To standardize input duration for model training, each curated audio track was segmented into non-overlapping 1-second intervals using the `pydub` library [21]. Let $x(t)$ denote the continuous-time audio waveform. The digital recording was divided into contiguous segments of fixed duration:

$$T_{\text{segment}} = 1 \text{ second.} \quad (1)$$

For an audio file of total duration D milliseconds, segmentation yields

$$N = \left\lfloor \frac{D}{1000} \right\rfloor \quad (2)$$

segments $\{x_0, x_1, \dots, x_{N-1}\}$.

Any residual segment shorter than one second was discarded to maintain consistent input dimensionality across all samples. Each extracted segment was exported as an independent `.mp3` file to facilitate modular data loading and reproducibility.

2.1.3. Label Encoding and Metadata Construction

Each audio track was associated with a predefined semantic label (e.g., *torpedo*, *whale*, *ships*). A categorical mapping dictionary was defined to convert string labels into integer class indices for supervised training: `class_dict = {whale : 0, fish : 1, unidentified : 2, physical : 3, ships : 4, communications : 5, sonar : 6, torpedo : 7}`. For each segment, the following metadata were recorded:

- Segment filename
- Original class label (string)
- Encoded class index (integer)

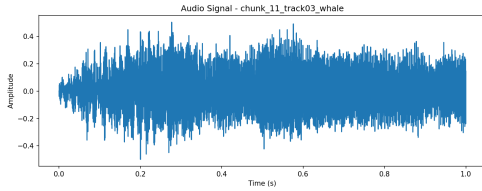
All metadata entries were stored in a structured CSV file (`labels_metadata.csv`) to support reproducibility and downstream data loading.

2.1.4. Dataset Statistics and Class Distribution

After manual curation and fixed-length segmentation, the final dataset comprised 1,099 non-overlapping 1-second audio segments across eight predefined acoustic classes. This segmentation strategy ensures consistent input dimensionality, reduced label ambiguity, and compatibility with batch-based

Table 1: Class distribution of the curated underwater dataset.

Class ID	Label	# Segments	Percentage (%)
0	Whale	273	24.84
1	Fish	99	9.01
2	Unidentified	94	8.55
3	Physical	122	11.10
4	Ships	81	7.37
5	Communications	200	18.20
6	Sonar	178	16.20
7	Torpedo	52	4.73

**Fig. 2:** Example waveform of a 1-second audio segment.

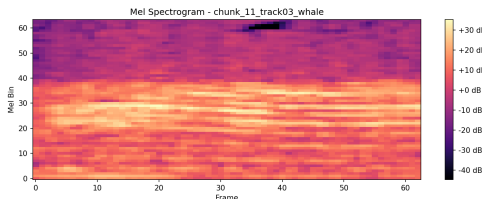
ML pipelines for supervised training and evaluation. The class distribution is moderately imbalanced, with the largest class containing 273 samples (24.84%) and the smallest class containing 52 samples (4.73%), resulting in an imbalance ratio (max/min) of 5.25. Such imbalance reflects the natural occurrence variability of underwater acoustic events and is considered during model training and evaluation. The detailed class distribution is summarized in Table 1.

2.2. Baseline Benchmark

To demonstrate the dataset’s utility and provide a reproducible benchmark, we train a simple Tiny-CNN for 8-class acoustic classification, reducing the risk of overfitting given the dataset size. The architecture is summarized in Table 2.

2.2.1. Feature Extraction

Each 1-second waveform segment is converted to a log Mel-spectrogram using a Mel-scaled STFT with $n_{\text{fft}} = 1024$, hop length = 256, sampling rate = 16,000 Hz, and $n_{\text{mels}} = 64$, following [22, 23].

**Fig. 3:** Mel-spectrogram representation of the corresponding 1-second audio segment.**Table 2:** Tiny-CNN architecture.

Layer	Configuration	Output Channels
Conv2D + BN + ReLU	3×3 kernel	32
MaxPool2D	2×2	32
Conv2D + BN + ReLU	3×3 kernel	64
MaxPool2D	2×2	64
Conv2D + BN + ReLU	3×3 kernel	128
Global Avg Pool	1×1	128
Fully Connected	Linear	8 classes

For a waveform $x(t)$ (Figure 2), the Mel-spectrogram (Figure 3) is computed as:

$$\mathbf{M} = \text{MelSpectrogram}(x; n_{\text{fft}} = 1024, \text{hop} = 256), \quad (3)$$

followed by logarithmic amplitude scaling:

$$\mathbf{M}_{\text{dB}} = 10 \log_{10}(\mathbf{M}), \quad (4)$$

implemented using an amplitude-to-decibel transformation with the dynamic range clipped to $[-80, 80]$ dB. To improve training stability and reduce sample-level energy variation, each log Mel-spectrogram is standardized as:

$$\hat{\mathbf{M}} = \frac{\mathbf{M}_{\text{dB}} - \mu}{\sigma + \epsilon}, \quad (5)$$

where μ and σ are the mean and standard deviation of the sample, and $\epsilon = 10^{-6}$ ensures numerical stability. The normalized Mel-spectrograms are then used as input to the Tiny-CNN.

2.2.2. Training Setup

The dataset is randomly split into 716 training, 164 validation, and 219 test samples, for a total of 1,099 segments. This split is fixed across all experiments to ensure fair comparison and reproducibility. The model is trained with cross-entropy loss and early stopping based on validation performance. Deterministic training is used for reproducibility, and all experiments are conducted on an NVIDIA GeForce RTX 3090 GPU and an AMD Ryzen Threadripper PRO 3955WX CPU. Detailed classification results are reported in Table 3, where the Support column indicates the number of test samples.

2.3. Cross-Domain Evaluation on ShipsEar

To evaluate cross-domain generalization, the pretrained 8-class model is tested on ShipsEar [20], a dataset of vessel and ambient noise recordings collected near the port of Vigo on the Spanish Atlantic coast. ShipsEar contains 90 recordings spanning 12 classes, including 11 vessel categories and one natural ambient-noise class. For consistency, all recordings are segmented into non-overlapping 1-second clips. The

Table 3: Baseline Tiny-CNN performance on the test set.

Class	Support	Precision	Recall	F1
Whale	61	0.967	0.951	0.959
Fish	13	1.000	1.000	1.000
Unidentified	16	0.941	1.000	0.970
Physical	21	0.909	0.952	0.930
Ships	14	1.000	0.857	0.923
Communications	40	0.927	0.950	0.938
Sonar	42	1.000	1.000	1.000
Torpedo	12	1.000	1.000	1.000
Macro Avg	219	0.968	0.964	0.965
Weighted Avg	219	0.964	0.963	0.963
Overall Accuracy	219		0.9635	

original labels are consolidated into a binary task: *ships* versus *ambient noise*. Predictions from the pretrained 8-class model are mapped to this 2-class setting by treating class ID 4 (Ships) as the positive ship class.

The evaluation set is highly class-imbalanced, comprising 11,300 segments, including 10,160 ship segments and only 1,140 ambient-noise segments. This imbalance poses a significant challenge, particularly for models not designed to handle skewed class distributions. As shown in Table 4, the baseline model trained with standard cross-entropy exhibits poor cross-domain performance, achieving only 5.91% ship detection rate and 15.31% overall accuracy. This degradation is primarily attributed to the combined effects of severe class imbalance and domain shift between the source training data and the ShipsEar dataset. Consequently, the model tends to favor more familiar source-domain classes, leading to frequent misclassification of ship segments despite their majority presence in the evaluation set.

2.4. Margin-Enhanced Loss and Feature Alignment for Class Confusion Mitigation

To improve robustness to both class imbalance and domain shift, we introduce two complementary components: a margin-enhanced loss during training and a lightweight feature-statistics alignment step during inference.

We incorporate class-dependent weighting into the cross-entropy loss [24] to reduce class imbalance. Let n_i denote the number of samples in class i . Following an inverse-frequency strategy, the class weights are defined as

$$w_i = \frac{\frac{1}{n_i}}{\frac{1}{C} \sum_{j=1}^C \frac{1}{n_j}} \quad (6)$$

where C is the number of classes. This normalization preserves an average weight of one while increasing the contribution of minority classes, thereby reducing bias toward dominant classes such as whale and sonar. Results show that ship samples (class 4) are frequently confused with whale (class

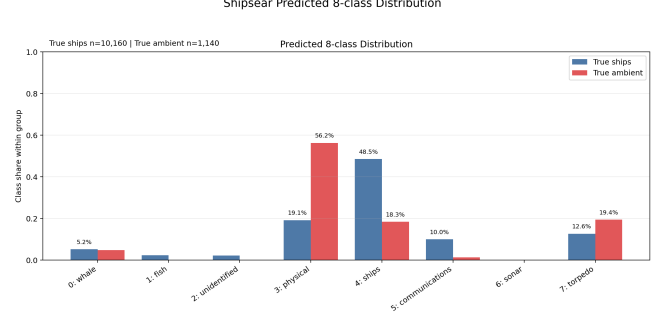


Fig. 4: ShipsEar classification report illustrating how ship and ambient noise being classified.

0) and sonar (class 6), primarily due to class imbalance and spectral similarity.

We introduce *CE-PlusPairMargin*, which augments the weighted cross-entropy with margin-based logit constraints inspired by the Adaptive-Weighted-Loss of Roy et al. [25]. For ship samples, we enforce a minimum margin between the ship logit and the competing whale and sonar logits:

$$z_4 - z_0 \geq m, \quad z_4 - z_6 \geq m \quad (7)$$

where z_i denotes the logit for class i and m is the margin parameter. The final loss is

$$\mathcal{L} = \mathcal{L}_{CE}(\mathbf{w}) + \lambda \mathcal{L}_{margin} \quad (8)$$

where $\mathcal{L}_{CE}(\mathbf{w})$ is the class-weighted cross-entropy and λ controls the margin regularization strength.

Because cross-domain degradation also arises from feature mismatch, we further apply a feature-statistics alignment step at inference time. Let $\mathbf{X}$ denote the target-domain log-mel spectrogram. Using the source-domain statistics (μ_s, σ_s) and target-domain statistics (μ_t, σ_t) , the aligned representation $\tilde{\mathbf{X}}$ is computed as

$$\tilde{\mathbf{X}} = \left(\frac{\mathbf{X} - \mu_t}{\sigma_t + \epsilon} \right) \sigma_s + \mu_s \quad (9)$$

where ϵ is a small constant for numerical stability. The aligned and original features are blended using an interpolation parameter α :

$$\mathbf{X}_{align} = (1 - \alpha)\mathbf{X} + \alpha\tilde{\mathbf{X}} \quad (10)$$

To avoid excessive shifts, the aligned features are clipped around the source statistics:

$$\mathbf{X}_{align} = \text{clip}(\mathbf{X}_{align}, \mu_s - k\sigma_s, \mu_s + k\sigma_s) \quad (11)$$

Finally, per-sample normalization is applied:

$$\mathbf{X}_{norm} = \frac{\mathbf{X}_{align} - \mu(\mathbf{X}_{align})}{\sigma(\mathbf{X}_{align}) + \epsilon} \quad (12)$$

Table 4: Cross-domain ship detection performance on ShipsEar - zero-shot and test-time feature alignment applied.

Model Tiny-CNN	Ship Det. Rate (%)	Ship F1	Overall Acc. (%)	Balanced Acc. (%)	# Segments
Standard CE zero-shot	5.91	0.111	15.31	52.51	11,300
Class-weighted CE	16.22	0.277	23.68	53.19	11,300
CE-PlusPairMargin	29.43	0.448	34.74	55.77	11,300
CE-PlusPairMargin + Feature Alignment	48.51	0.644	51.86	65.09	11,300

This alignment acts as a lightweight domain adaptation mechanism and requires no retraining. While the margin-enhanced loss yields limited gain on the near-saturated 8-class in-domain task, it substantially improves cross-domain ship detection. As shown in Table 4, ship detection increases from 5.91% under standard CE to 29.43% with margin-enhanced training, and further to 48.51% after feature alignment as shown in Table 4 and Figure 4.

3. RESULTS AND ANALYSIS

3.1. Evaluation Metrics

Let TP , TN , FP , and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

Ship detection rate is defined as the recall of the ship class:

$$\text{Recall}_{\text{ship}} = \frac{TP}{TP + FN}. \quad (13)$$

The F1 score combines precision and recall:

$$F_1 = \frac{2PR}{P + R}, \quad (14)$$

where

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}. \quad (15)$$

Balanced accuracy accounts for class imbalance by averaging recall across classes:

$$\text{Balanced Accuracy} = \frac{\text{Recall}_{\text{ship}} + \text{Recall}_{\text{ambient}}}{2}. \quad (16)$$

Overall accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (17)$$

3.2. In-Domain Performance Analysis

Table 3 summarizes in-domain results on the USS8 test set. The Tiny-CNN achieves 96.35% overall accuracy with macro and weighted F1-scores of 0.965 and 0.963, indicating stable performance despite dataset imbalance. Perfect precision and recall are observed for *Fish*, *Sonar*, and *Torpedo*, suggesting

distinct spectral characteristics. In contrast, the *Ships* class achieves perfect precision (1.000) but lower recall (0.857), indicating that some ship samples are confused with acoustically similar categories. Slightly lower F1-scores for *Physical* and *Communications* also reflect residual inter-class overlap. Overall, the results confirm that the curated dataset is learnable with a lightweight architecture while revealing early signs of ship-class confusion.

3.3. Domain Shift Analysis

Despite strong in-domain performance, cross-domain evaluation under standard CE training shows severe degradation, indicating substantial distribution shift between USS8 and ShipsEar. Differences in recording hardware, environmental conditions, vessel operating modes, and background noise profiles introduce spectral and temporal mismatch. In addition, USS8 contains fewer vessel types than ShipsEar, which constrains the model’s ability to learn sufficiently diverse and robust vessel-specific features, thereby further increasing domain discrepancy. To analyze the impact of feature alignment, we compare predictions before and after alignment. Among the 11,300 evaluated segments, 8,802 predictions (77.89%) change class labels, indicating that the alignment significantly reshapes the feature distribution perceived by the classifier. The most frequent prediction transitions are summarized as:

$$\text{TopTransitions} = \begin{cases} 0 \rightarrow 4 : 2347 \\ 0 \rightarrow 3 : 1542 \\ 0 \rightarrow 7 : 746 \\ 4 \rightarrow 3 : 520 \\ 0 \rightarrow 5 : 486 \\ 3 \rightarrow 5 : 373 \\ 2 \rightarrow 4 : 351 \\ 4 \rightarrow 7 : 301 \\ 6 \rightarrow 4 : 295 \\ 3 \rightarrow 7 : 267 \end{cases} \quad (18)$$

Inspection of class transitions shows that the most frequent correction is whale-to-ship ($0 \rightarrow 4$), affecting 2,347 segments. This directly addresses the dominant confusion pattern from the results analysis. Additional transitions such as $2 \rightarrow 4$ (351 segments) and $6 \rightarrow 4$ (295 segments) further

contribute to improved ship detection. Other frequent changes include $0 \rightarrow 3$ (1,542 segments) and $0 \rightarrow 7$ (746 segments), indicating that the alignment globally adjusts the feature distribution rather than targeting a single class boundary.

3.4. Discussion

Despite achieving 96.35% in-domain accuracy, the CE-trained model detects only 5.91% of ships in the cross-domain setting, showing that strong closed-set performance does not necessarily translate to generalization. The margin-enhanced loss and feature alignment improve cross-domain detection to 48.51 %, indicating that structured inter-class separation improves robustness to acoustically overlapping categories. However, the remaining gap suggests these methods alone cannot fully address domain mismatch. Moreover, although computationally efficient, the Tiny-CNN may lack sufficient capacity to model the complex temporal dynamics present in real-world maritime recordings.

4. CONCLUSIONS

Despite the improvements achieved with margin-enhanced training, limitations still remain. Differences in vessel characteristics, operational behavior, and environmental conditions are not fully represented in the current dataset, limiting cross-domain generalization. In addition, the transfer evaluation relies on direct model reuse without explicit domain adaptation or feature alignment. As reflected in the cross-domain performance gap, distribution mismatch between datasets remains a major challenge. Future work will focus on expanding ship diversity in the dataset, incorporating domain adaptation strategies such as feature alignment or limited target-domain fine-tuning, and exploring self-supervised or contrastive pretraining to learn more transferable acoustic representations. These efforts aim to improve robustness and support more reliable deployment in real-world underwater surveillance scenarios.

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